

Given:

Each result column $[r_1, r_2, r_3, r_4], [r_5, r_6, r_7, r_8], \dots$ is computed by multiplying the **MixColumns matrix** with each **column** of the state matrix:

MixColumns matrix:

```
[02 03 01 01]
[01 02 03 01]
[01 01 02 03]
[03 01 01 02]
```

State matrix (ShiftRows output):

```
[B3 47 A2 F0]
[9C 63 C5 F2]
[7B 7C F0 AB]
[CA AF 76 76]
```

◆ First Column (B3, 9C, 7B, CA)

$$\begin{aligned}r_1 &= (02 \times B3) \oplus (03 \times 9C) \oplus (01 \times 7B) \oplus (01 \times CA) \\r_2 &= (01 \times B3) \oplus (02 \times 9C) \oplus (03 \times 7B) \oplus (01 \times CA) \\r_3 &= (01 \times B3) \oplus (01 \times 9C) \oplus (02 \times 7B) \oplus (03 \times CA) \\r_4 &= (03 \times B3) \oplus (01 \times 9C) \oplus (01 \times 7B) \oplus (02 \times CA)\end{aligned}$$

◆ Second Column (47, 63, 7C, AF)

$$\begin{aligned}r_5 &= (02 \times 47) \oplus (03 \times 63) \oplus (01 \times 7C) \oplus (01 \times AF) \\r_6 &= (01 \times 47) \oplus (02 \times 63) \oplus (03 \times 7C) \oplus (01 \times AF) \\r_7 &= (01 \times 47) \oplus (01 \times 63) \oplus (02 \times 7C) \oplus (03 \times AF) \\r_8 &= (03 \times 47) \oplus (01 \times 63) \oplus (01 \times 7C) \oplus (02 \times AF)\end{aligned}$$

◆ Third Column (A2, C5, F0, 76)

$$\begin{aligned}r_9 &= (02 \times A2) \oplus (03 \times C5) \oplus (01 \times F0) \oplus (01 \times 76) \\r_{10} &= (01 \times A2) \oplus (02 \times C5) \oplus (03 \times F0) \oplus (01 \times 76) \\r_{11} &= (01 \times A2) \oplus (01 \times C5) \oplus (02 \times F0) \oplus (03 \times 76) \\r_{12} &= (03 \times A2) \oplus (01 \times C5) \oplus (01 \times F0) \oplus (02 \times 76)\end{aligned}$$

◆ Fourth Column (F0, F2, AB, 76)

$$\begin{aligned}r_{13} &= (02 \times F0) \oplus (03 \times F2) \oplus (01 \times AB) \oplus (01 \times 76) \\r_{14} &= (01 \times F0) \oplus (02 \times F2) \oplus (03 \times AB) \oplus (01 \times 76) \\r_{15} &= (01 \times F0) \oplus (01 \times F2) \oplus (02 \times AB) \oplus (03 \times 76) \\r_{16} &= (03 \times F0) \oplus (01 \times F2) \oplus (01 \times AB) \oplus (02 \times 76)\end{aligned}$$

These are the exact equations you'll use for calculating r_1 to r_{16} during the AES MixColumns step. If you want me to **solve these with actual hex results**, let me know — I can give full answers using Python or manual steps.